DO NOW — RECALL

Writing, following and documenting algorithms

Name: _____ Class: _____ Date: _____

1. Give a test for a range 0–100 that should be rejected.

2. What does return do?

3. What is the last index for an array of length 4?

4. Why is a[0] unsafe for an empty array?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

−1 or 101.

2. Model answer

Passes a function's result back to the caller.

3. Model answer

3.

4. Model answer

Index 0 does not exist, causing an out-of-range error.

Vocabulary — know these terms

algorithm — Precise finite steps that solve a problem.

precondition — Requirement inputs must meet before execution.

postcondition — Condition guaranteed if the algorithm completes correctly.

invariant — Property maintained during repeated steps.

documentation — Clear description of purpose, inputs and expected behaviour.

NEW LEARNING

Read, annotate and explain

State the precondition

The array must not be empty, or a[0] cannot be read.

Trace the maximum

Set best to 3; compare 7, update best to 7, then ignore 2 and 5.

Document contract

Input: nonempty int array. Output: greatest element. Does not modify input.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace Largest([4,2,9]) and write the returned result.

Predict first

Start best at first element 4; 2 does not replace it.

Model and explain

9 is larger so best becomes 9; return 9.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace Largest([2,5,1]) and state initial and final best.

Question 2

Write pseudocode to return the minimum of a nonempty array values and explain its precondition.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate what happens for Largest([-4,-2,-9]) and explain why initialising best=0 would be incorrect.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Initialising a maximum-tracking variable to 0 is correct for every integer array.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.